

A Robotic Wheel Locally Transforming the Diameter for the Locomotion on Rough Terrain

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Abstract - A locomotion ability of robots has been studied to move on a various terrain stably. We are developing a robotic wheel which locally transforms its diameter to improve the mobility on an uneven terrain. The robotic wheel locomotes on a flat road using the high mobility of a wheel, and also on rough terrains by changing its diameters. We introduce a variable diameter wheel as a prototype of the robotic wheel, and investigate its availability. Without installing an axle motor, the robot is able to locomote by utilizing the extension and returning of the diameters by moving its centre of gravity. The extension of the spokes works to push the robotic body in order to locomote, and the returning of the spokes make the wheel get the rotation momentum. The extension and returning mechanism is driven by rack & pinion mechanism. We verified its mobility through four experiments, which are the testing locomotion on flat ground, the climbing over the step, controlling the robotic wheel by IMU, and the braking by the extension of spokes. The trajectory which the tip of the spoke draws is also compared with the simulation using MATLAB.

Index Terms – Robotic wheel, Variable diameter, Climbing over obstacles.

I. INTRODUCTION

There are generally two types of robotic locomotion introduced for the robust mobility on different types of grounds. One is to use its legs and the other is to employ its wheel. The former is suitable for the efficient locomotion on a rough ground, and the latter gives better performance on a flat ground and a slope.

In this research, we pay attention to the moving ability of a wheel, and aim to introduce the new functional mobility of the robotic wheel locomoting on a rough terrain. Fig. 1 shows the concept of climbing over obstacle together with the picture of our robotic wheel.

Different robots with a specific wheel to climb the obstacles have been introduced so far. For example, Tadakuma et al. developed a robot with spherical omni wheels that can move all directions and even on uneven terrains [1]. Herbert et al. designed a robotic wheel shaped like a boomerang [2]. This robot has high locomoting ability on rough terrains without having variable wheels. Alsaman et al. simulated a wheel that could locomote on rough terrain without deforming its wheel actively but had a passive mechanism [3]. Through the simulation, an optimal plane trajectory was calculated, and the disturbance while locomoting could be reduced. Chen et al. introduced a robotic wheel having a system to change its wheel into legs [4] [5].

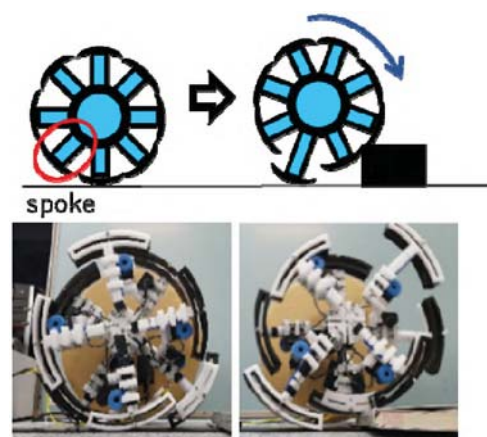


Fig. 1 Concept of the robotic wheel for locomotion and climbing over an obstacle (top), pictures of an assembled robotic wheel (bottom).

This robotic wheel can transform its wheeled form into a legged mode by folding the wheel to a half moon shape. Zheng et al. have studied a robotic wheel which can change its wheel to a boomerang-like shape [6]. Since the variable part has a nail-like shape, it is easily hooked on a step and has high climbing performance. Koh et al. studied a robot that could change the diameter of wheels using shape memory alloys, flexure linkages, and deformable polymers [7]. By changing the ratio between the diameter of the wheel and the height of the step, the climbing performance is improved. Kim et al. developed a wheel with a mechanism that was passively deformed to omit the actuator [8]. When the wheel contacts a step, it is passively deformed into a boomerang-like shape. She et al. have also developed a wheel with a passive mechanism that transforms into a pentagon with a few actuators [9]. Without using the actuators for all of the deformation, the load on the wheel is reduced. Jeans et al. studied a robot that has no rims but can move over rough terrain by moving spokes [10]. It consists of two wheels with a tail, and it can climb over a higher step. Felton et al. invented an origami wheel that could passively change its radius depending on the load [11]. Lee et al. have developed an origami wheel transformer that can change its diameter using SMA coil spring actuators [12]. Owing to the deformation of the wheel diameter, the wheel can locomote through the small gaps. They also developed wheels with a different origami structure [13]. Chen et al. studied a mobile wheelchair robot for all terrains, especially for staircases and inclines. This robot can get on and off steps and slopes while keeping its

body horizontal [14]. Xiaojiang et al. studied a variable diameter wheel, H-drive four-wheel drive system and its transmission ratio. The study shows that the grade ability is inversely proportional to the diameter of the wheels therefore the wheel can locomote different slopes optimally [15].

As described above, many researchers have studied the wheels which transform entirely or have a special structure. However, no attention has been paid to the wheels having a mechanism capable of locally deforming.

We propose a novel robotic wheel, which locally transforms its diameter to solve the above problems. Our robotic wheel can move efficiently on the flat road by using its own ability of wheel, and gets over obstacles by transforming its diameter locally or entirely. In this paper, we firstly introduce the fundamental design of the robotic wheel, and then describe the prototype of a wheel transformable robot. The trajectory of the tip of the spoke is compared with the simulation using MATLAB. Four experiments are performed to verify the locomotion performance of wheel on flat ground, the ability to climb over a step, feedback control using an IMU sensor, and the braking performance of the wheel.

II. FUNDAMENTAL DESIGN OF A ROBOTIC WHEEL

A. Requirement

1) Design Requirement for a Robotic Wheel

A wheel in general has a frame structure by spokes inside. In this research, we propose a transformable robotic wheel by dividing the wheel equally and radially. The rim is divided into eight parts by considering the symmetry of the wheel and the constraint of the size. Each part of the rim is connected to a spoke, which linearly extends and returns the diameter for the locomotion.

We consider four requirements to realize the local transformation of the wheel, which are (1) the size of the wheel, (2) the mechanism of local and independent transformation of the wheel, (3) the power to support the body for locomotion, and (4) the control of the extension and the returning of a spoke.

To meet the requirements (1) and (2), we adopt a rack & pinion mechanism, which is low cost and compact structure. This mechanism also has an advantage for transferring the power efficiently from the motor torque to the pinion gear. By implementing this mechanism into each spoke, the diameter of the wheel can be controlled independently. The requirements (3) and (4) are met by using servo motors for each spoke because they have the sufficient torque and can control spokes accurately.

2) Requirements for Servo Motor

To select the motors for driving the rack & pinion mechanism, we estimate the required torque for the spoke articulation. To lift the entire wheel by the spoke articulation, motors in charge are required to generate sufficient torque. The torque requirement for each spoke is calculated by considering the geometry of the wheel, which is described in chapter III.

Here, we estimate the maximum and minimum torque for the wheel locomotion. The total weight of the robot body is

4kgf and the radius of the pinion gear is 1.3cm. The minimum required torque is estimated to be 5.2kgf-cm, and the servo motor is selected to meet the requirement. The specification of the servo motor (ROBOT SERVO RS301CRF3 FUTABA) is shown in Table 1.

TABLE I
SPECIFICATION OF SERVO MOTOR

torque	7.1 kgf-cm
speed	0.11sec/60°
mass	28g

B. Mechanism and Concepts

1) Mechanical Design of the Wheel

We design the robotic wheel by using a CAD software. The design and appearance of the entire body is shown in Fig. 2. The foundation is designed with a rail structure that allows the spokes to slide stably as shown in Fig. 3.

The picture of the assembled robotic wheel based on the design drawing is shown in Fig. 1. The robotic parts are printed by a 3D printer (AGILISTA-3200, KEYENCE) and also fabricated by a modeling plotter (NC-5SK, Mimaki). The materials of the parts are acrylate resin and ABS.

2) Rack & Pinion Mechanism

A pinion gear connected with a servo motor is combined with a rack gear with a spoke as shown in Fig. 4. The extension can be achieved by the rack & pinion mechanism up to the half length of a radius. Each spoke moves in the direction of extension from the initial radius of the wheel, and



Fig. 2 Design of a robotic wheel.

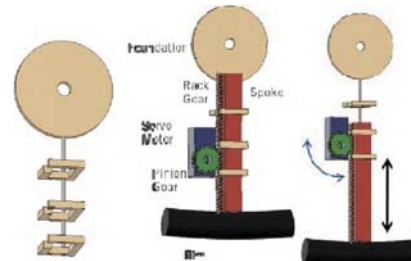


Fig. 3 Mechanism for extension and returning of the wheel.

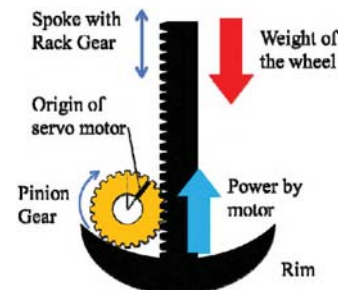


Fig. 4 Mechanism of the spoke.

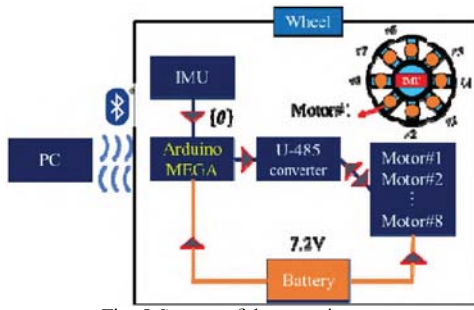


Fig. 5 System of the experiments

returns to the initial length. A pinion gear is attached to the shaft of a servo motor, and the rotating motion is transmitted to a rack gear with the spoke. The rotation of the pinion gear is controlled by a servomotor, and the spoke extends by following clockwise rotation from the origin of rotation.

3) Locomotion and Climbing over an Obstacle

Without using axle motors, our robotic wheel can independently locomote on a flat road and climb over an obstacle by controlling the extension and returning timing of each spoke to move its center of the gravity. An extending action of a spoke pushes a ground for starting and continuing the locomotion. When the wheel contacts with an obstacle, the moving direction of the wheel goes up to climb over the obstacle. The spoke contacting with the ground is extended at the same time, and the spoke contacting to the obstacle contracts, so that the wheel climbs the obstacle.

C. System of the Robotic Wheel

Fig. 5 shows the system of the robotic wheel. All motors of the robotic wheel are controlled by Arduino. The control signals are converted using a U-485 converter to be sent from USB to RS485 for establishing serial communication between Arduino and the motors. The servo motors send angle data back to Arduino. Arduino and a PC are connected with each other using Bluetooth. A small battery supplies 7.2V to the motors, Arduino main unit and each control module. To control the wheel motors, one IMU (InvenSense MPU 9250) sensor is installed near the center of the wheel for obtaining the rotation angle of the robotic body. It is expected to control the speed of the robotic wheel and improve its operability. The IMU is controlled by performing I2C communication using SDA and SCL pins. The rotation angle of the robotic wheel can be calculated by inputting the angular acceleration of the wheel into the Kalman filter stored in Arduino, and then Arduino sends the commands of angle data back to the servo motors as a feedback. In experiment 1 and 2, the IMU sensor is not used, and the experiments are performed as feedforward control.

III. MOTION MODELING

Before conducting experiments using actual robotic wheel, some parameters are determined by modeling the trajectory of the tip of a spoke. With the use of the determined parameters, an actual trajectory is expected to be similar to the modeled trajectory. The trajectory of the transformable robotic wheel was modeled by using MATLAB. In the stationary state

of the wheel, the coordinates of the tip of the spoke to be driven first are set on the y -axis. Each spoke is numbered counterclockwise as shown in Fig. 6. Here, we suppose a case that spoke #1 is located at a position where the spoke rotates 40° to the left from the vertical direction. As far as the spoke is not positioned vertically downward, the wheel starts rotating by extending the spokes. The coordinates of the tip of the spoke #1 is set as $w(x, y)$. Here, $x = 0$ and y can be determined geometrically by considering the location of the wheel center $C_o(x, y)$, since the radius of the wheel is known. The y -coordinate of the tip of the spoke is given by $r - r \cos(40^\circ)$, where r is the radius of the wheel. The spoke #1 is extended to rotate the wheel. By extending the spokes from #1 to #4 in the numerical order, the robotic wheel locomotes to the right direction. The trajectory of the coordinates $w(x, y)$ is simulated.

In general, when a wheel rotates without slipping, a point on the circumference of the wheel follows a cycloid trajectory. The cycloid on the x - y plane is given by the following equation using parameters r representing the radius of the circle that draws the cycloid and θ representing the rotation angle of the circle.

$$x(r, \theta) = r \cdot (\theta - \sin(\theta)) \quad (1)$$

$$y(r, \theta) = r \cdot (1 - \cos(\theta)) \quad (2)$$

Furthermore, the transformable robotic wheel rotates by extending and returning its spokes, and an additional term needs to be considered as shown in the following equations.

$$x(r, \theta, t) = r \cdot (\theta - \sin(\theta)) \pm (1.3 \cdot a \cdot \pi / 180) \cdot t \cdot \sin(\theta) \quad (3)$$

$$y(r, \theta, t) = r \cdot (1 - \cos(\theta)) \pm (1.3 \cdot a \cdot \pi / 180) \cdot t \cdot \cos(\theta) \quad (4)$$

Here we note that r and θ are the same parameters with a normal cycloid. In addition, a represents the angular velocity of the pinion gear and t represents the time required for extension and returning. We empirically chose the angular velocity of the motor as 360 deg/sec. Fig. 7 shows the

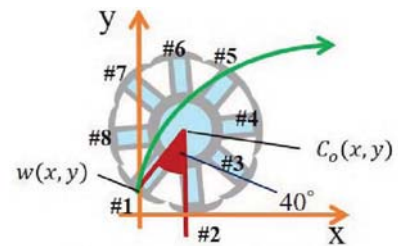


Fig. 6 Setting of the simulation.

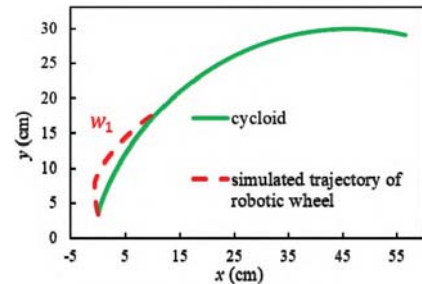


Fig. 7 Modeling of trajectory.

comparison of the normal cycloid with the trajectory of the robotic wheel.

IV. LOCOMOTION EXPERIMENT OF FEEDFORWARD CONTROL

A. Locomotion on Flat Surface (Experiment 1)

We firstly verified the mobility of the robot by only using the extension and returning mechanism. The relation between the angular velocity and the extension/returning time of the motors shown in (3) and (4) are studied. The spokes are extended in numerical order as presented in a signal timing chart shown in Fig. 8. The axis of the ordinate shows the displacement angles of the motor. The positive slope corresponds to the rotational direction for the extension, and on the other hand the negative slope presents the contraction of the spoke. The angular velocity of the servo motor corresponding to the parameter a is given by 360 deg/sec, which corresponds to the slope of the graph. The extension and returning time of the motor is all set with 0.5 seconds. By returning the extended spokes to the contracting directions, the robotic wheel is able to gain angular momentum to move the barycenter of the wheel forward. Owing to the acceleration that occurs when the spoke is returned, a force in the direction of acceleration is generated in the wheel body. A propulsion is generated by the vector component of the force in the traveling direction of the wheel, and the momentum is obtained. We observed that the robotic wheel could move smoothly as shown in Fig. 9, where the pictures were taken in every 0.6 seconds. To clearly show the rotating wheel in the running, one spoke is surrounded by a red circle.

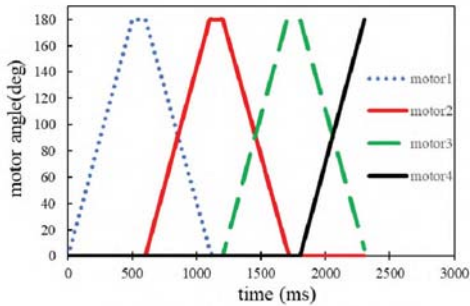


Fig. 8 Timing chart of Experiment 1.

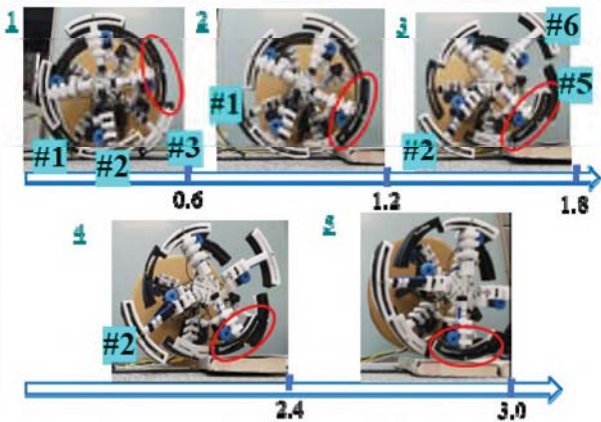


Fig. 9 Time series of locomotion of the robotic wheel.

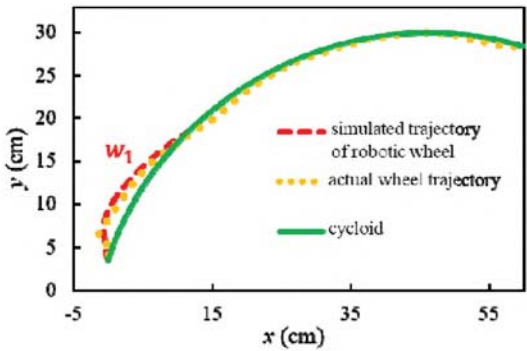


Fig. 10 Comparison of three trajectories.

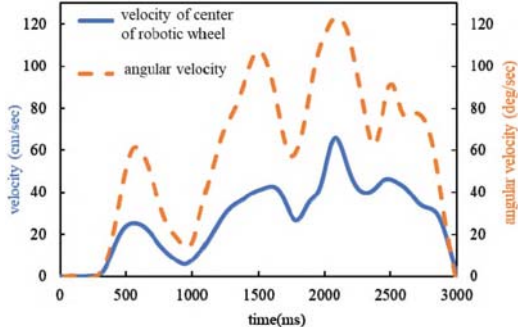


Fig. 11 Change of velocity and angular velocity in Experiment 1.

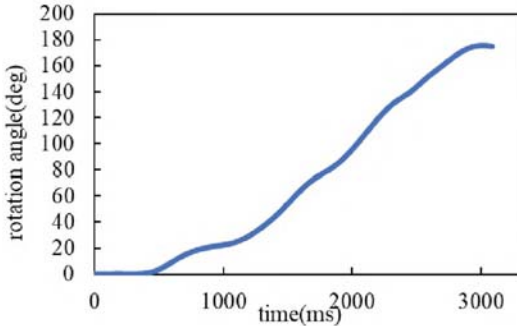


Fig. 12 Rotational angle in locomotion in Experiment 1.

We compare the simulated trajectory with the actual wheel trajectory. The coordinate of the tip of the initially-driven spoke is measured by a software (Kinovea) using a video. As shown in Fig. 10, the actual trajectory of the robotic wheel reproduced the simulated one properly. In the actual trajectory at the beginning, it does not match with the simulated trajectory because the wheel does not instantly start locomoting as the spoke is extended, due to the inertia of the wheel itself. However, after a couple of seconds, the locomotion matches with the simulation, which validates the introduced mechanism for the locomotion.

We also analyze the mobility in the experiment. The velocity and the angular velocity of the wheel during the locomotion are calculated by using the video frames, which are shown in Fig. 11. Fig. 12 also presents the rotation angle of the wheel. The velocity changes according to the pushing actions of spokes to generate acceleration, however the wheel could locomote rather smoothly by the articulation of plural spokes.

B. Verification of Obstacle Climbing (Experiment 2)

We examine the ability to climb over an obstacle. For climbing over the obstacle, it is necessary to lift the robotic body by extending the spoke to move the barycenter of the robotic wheel over the obstacle. In this experiment, a step with the height of 2.10cm is located at a distance of 3.00 cm in front of the robotic wheel. We heuristically analyzed the articulation of the spokes, and tried a possible control method for the climbing, which is shown in Fig. 13. The speed of the wheel during the climbing is calculated by using the video frames, which is shown in Fig. 14. Fig. 15 also presents the rotation angle of the wheel.

First, spoke #1 is extended, and simultaneously #3 contracts to rotate the wheel. When the wheel reaches the step,

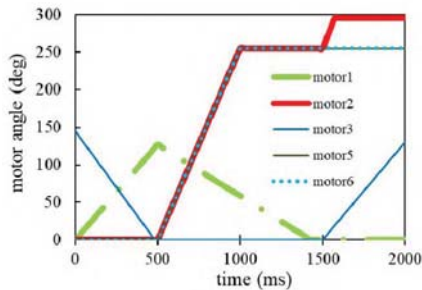


Fig. 13 Timing chart of motor angle in Experiment 2.

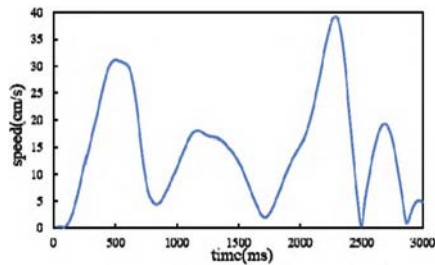


Fig. 14 Change of the speed in Experiment 2.

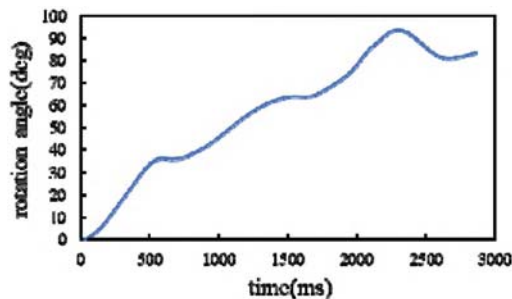


Fig. 15 Rotation angle in climbing in Experiment 2.

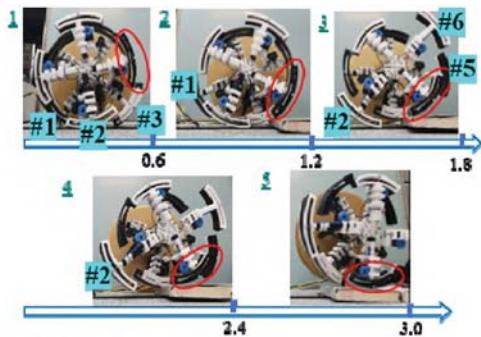


Fig. 16 Scene of motion in climbing over a step.

spoke #2 is extended to push the ground and raise the body. At the same time, spokes #5 and #6 are extended to move the barycenter to the direction over the step. These sequential extensions and returning of the spokes made a successful motion for climbing over the step.

Fig. 16 shows the temporal locomotion of the wheel climbing over the step for every 0.6 seconds. The angular velocities of the spokes #1, #2, #3, #5, #6 are 255, 510, 300, 510, 510 deg/sec. The initial positions of each motor are set at 0 degrees while #3 is set at 145 degrees.

V. LOCOMOTION EXPERIMENT OF FEEDBACK CONTROL

A. Feedback Control Using an IMU Sensor (Experiment 3)

In the previous sections, the timing for each motor is given in advance empirically. Here, we additionally implemented an IMU sensor to obtain the rotational angle of the wheel body for the feedback control. The IMU sensor outputs an inclinational angle of the body, and the spokes corresponding to the body angle are adaptively controlled. The relation between the body inclination and the corresponding spokes are experimentally established as shown in Table II. In this experiment, in order to let the wheel autonomously locomote, feedback control is performed by referring to the table. For example, when the wheel body is situated at the inclination of 71 degrees, the spoke #2 is extended and at the same time the spoke #1 is returned, so that the robotic body moves forward.

For the calculation of the body angle, Kalman filter is employed as shown in Fig. 17 for the smooth locomotion. Fig. 18 shows the transition of the rotational angle of the robotic body during the autonomous locomotion using the feedback control. Smooth rolling locomotion was achieved by the wheel robot.

B. Evaluation of the Braking Ability (Experiment 4)

The robot is able to stop its locomotion by articulating the spokes for braking. Braking performance can be exhibited only by extending and returning spokes without using brake pads. Spokes required for braking can be controlled according to the rotation angle of the wheel. The relation between the body angle and the corresponding spokes for braking is also

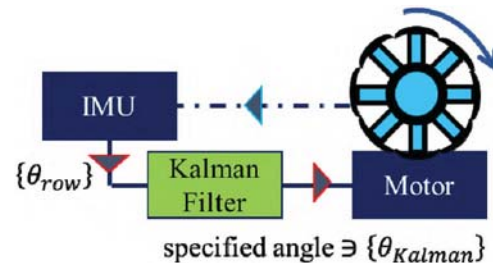


Fig. 17 Feedback control in Experiment 3.

TABLE II
SPOKES CORRESPONDING TO ROTATIONAL ANGLES FOR LOCOMOTION

Body angle [degrees]	Extending Spoke #	Returning Spoke #
10	1	-
26 ~ 28	1	-
71 ~ 73	2	1
116 ~ 118	3	2
161 ~ 163	4	3

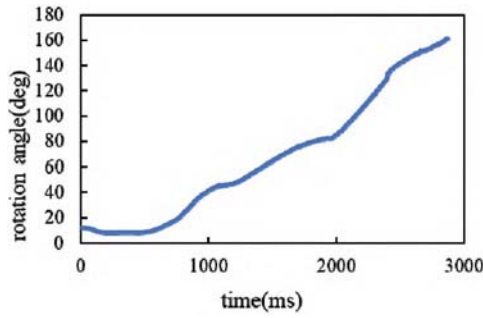


Fig. 18 Transition of the body rotational angle in locomotion.

TABLE III

SPOKES CORRESPONDING TO ROTATIONAL ANGLES FOR BRAKING

Body angle [degrees]	Extending Spoke #	Returning Spoke #
10	1	-
26 ~ 28	1	-
71 ~ 73	2	1
116 ~ 118	3	2
161 ~ 163	4	3
206 ~ 208	5, 7	4

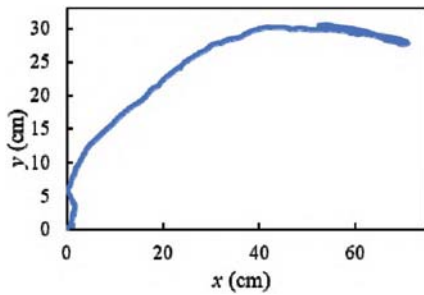


Fig. 19 Trajectory of the tip of spoke #1 in Experiment 4.

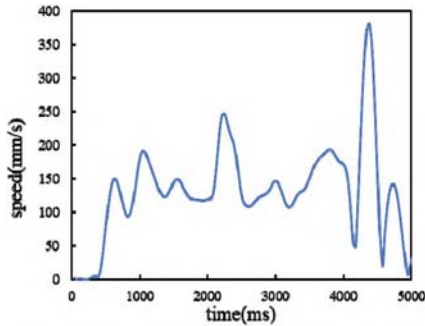


Fig. 20 Change of the speed in Experiment 4.

empirically determined as shown in Table III, and the braking is autonomously executed by obtaining the angles from the IMU sensor. Fig. 19 shows the braking results, in which the trajectory of the tip of spoke #1 is presented as an example. We could verify that the braking performance was successfully made by only extending and returning spokes. Fig. 20 shows the speed of the center of the robotic wheel.

VI. CONCLUSIONS

In this research, we developed a novel robotic wheel which locally transforms its diameter to locomote on uneven terrain. The robotic wheel was able not only to locomote on a flat surface but to climb over a step. By using the IMU sensor,

feedback control was performed to extend spokes according to the rotation of the wheel. Braking performance was also verified by the articulation of spokes. In the next study, the extension and returning mechanism would be improved to obtain the better ability of getting over the higher and much rough obstacles. We also plan to install a vision system to measure the situation of terrain for realizing the adaptive locomotion.

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